

Python Libraries Used

1. numpy
2. pandas

Python Calculated Fields

c_ThreatScore_CurrentYear

```
import numpy as np
```

```
loss = np.array(_arg1)
```

```
users = np.array(_arg2)
```

```
incidents = np.array(_arg3)
```

```
# Normalize with zero-division protection
```

```
def normalize(arr):
```

```
    min_val = arr.min()
```

```
    max_val = arr.max()
```

```
    return (arr - min_val) / (max_val - min_val) if max_val != min_val else np.zeros_like(arr)
```

```
loss_norm = normalize(loss)
```

```
users_norm = normalize(users)
```

```
incidents_norm = normalize(incidents)
```

```
# Weighted Threat Score
```

```
score = (0.5 * loss_norm + 0.3 * users_norm + 0.2 * incidents_norm) * 100
```

```
return score.tolist()
```

c_ThreatScore_PreviousYear

```
import numpy as np
```

```
loss = np.array(_arg1)
```

```
users = np.array(_arg2)
```

```
incidents = np.array(_arg3)
```

```
def normalize(arr):
```

```
    min_val = arr.min()
```

```
    max_val = arr.max()
```

```
    return (arr - min_val) / (max_val - min_val) if max_val != min_val else np.zeros_like(arr)
```

```
loss_norm = normalize(loss)
```

```
users_norm = normalize(users)
```

```
incidents_norm = normalize(incidents)
```

```
score = (0.5 * loss_norm + 0.3 * users_norm + 0.2 * incidents_norm) * 100
```

```
return score.tolist()
```

c_Arrows

```
result, lista = [], []
```

```
actu = _arg1
```

```
prev = _arg2
```

```
lista = [(i-j) if i!=None and j!=None else None for i,j in zip(actu,prev)]
```

```
for i in lista:
    if i == None:
        result.append("")
    elif i<0:
        result.append('▲')
    elif i>0:
        result.append('▼')
    else:
        result.append('▼ ▲')
return result
```

c_py_SecurityRank(1-10)

```
import pandas as pd
serie = pd.Series(_arg1)
rangos = serie.rank(method='min').astype(int)
return rangos.tolist()
```